

Dealing with small sample size problems in process industry using virtual sample generation: a Kriging-based approach

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Abstract

The operational data of advanced process systems have met with explosive growth, but its fluctuations are so slight that the number of the extracted representative samples is quite limited, making it difficult to reflect the nature of the process and to establish prediction models. In this study, inspired by the process of fisherman repairing nets, a Kriging-based virtual sample generation (VSG) named Kriging-VSG is proposed to generate feasible virtual samples in data sparse regions. Then, the accuracy of prediction models is further enhanced by applying the generated virtual samples. In order to reasonably find data sparse regions, a distance-based criterion is imposed on each dimension to identify important samples with large information gaps. Similar to the process of fisherman repairing nets, a certain dimension is initially fixed at different quantiles. A dimension-wise interpolation process using Kriging is then performed on the center between important samples with large information gaps. To validate the performance of the proposed Kriging-VSG, two numerical simulations and a real-world application from a cascade reaction process for high-density polyethylene are carried out. The results indicate that the proposed Kriging-VSG outperforms other methods.

Keywords Small sample size problems · Virtual sample generation · Kriging interpolation · Soft sensing modeling · High-density polyethylene

1 Introduction

Process modeling becomes exceedingly important to automatically handle process abnormalities and irregular events (Garg and Mhaskar 2018). Although the modern process industries equipped with lots of advanced

technology enable to record huge amounts of data from production operations, management behaviors and other actions, it is difficult to obtain enough representative samples for establishing models so that capture the intrinsic characteristics of the regular operation and potential abnormalities for process systems. On one hand, the coexist of small fluctuation among data, the high cost of data acquisition or the low probabilities of abnormal event occurrences greatly reduce the size of representative

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→ auto custo de aquisição de dados

samples. On the other hand, such data have strong non-linearity, noise, missing values and uncertainty, which makes data-driven models hard to extract information and knowledge effectively. The performance of data-driven models is seriously restricted by the insufficient volume, poor representation and imbalanced distribution of collected samples (Chen et al. 2017). Therefore, these challenges can be treated as small sample size problems regarding representative samples.

Small sample size problems are not exclusive to process industry, but prevalent in the fields of computer science (Zhu et al. 2019), biomedical engineering (Liu et al. 2019; Shaikhina et al. 2015) and material science (Feng et al. 2019; Zhang and Ling 2018) as well. As reported in the literature, many efforts have been made to overcome small sample size problems in the tasks of soft sensing modeling (Tang et al. 2017), fault diagnosis (Gao et al. 2019), energy prediction and analysis (He et al. 2018), pattern recognition (Zhu et al. 2019) and medical diagnosis (Shaikhina and Khovanova 2017; Shaikhina et al. 2015), shown in Fig. 1. A variety of well-established machine learning methods are used to directly learn on small datasets. These methods involve gray-based methods (Talafose and Pohl 2017), support vector machines (SVMs) (Dong et al. 2018; Hong et al. 2019; Liu et al. 2016), kernel regression (Shapiat et al. 2011) and Bayesian networks. A gray-based method is one of the most widely used tools to model and analyze an uncertain system in which part of information is known and part of information is unknown. It works well on single-dimension data that vary exponentially, but it is

difficult to apply to unevenly spaced time series data. SVM describes data distribution structurally by margin, which reduces the requirements on data size and distribution. In many engineering applications, SVMs have a better performance against traditional neural networks when training datasets are limited. However, it is quite sensitive to noisy data or outliers. Bayesian networks are acyclic directed graphs capturing the probabilistic dependencies and independencies among variables. The dominant advantage of Bayesian networks over other methods is that it is easy to combine exiting data with expert judgment within their probabilistic framework to carry out for reasoning with uncertainty. But obtaining network structure and parameters of Bayesian networks from small datasets is generally prohibitive. Though learning paradigms like transfer learning (Saha et al. 2015) and active leaning (Ge 2014) have been incorporated with machine leaning methods for solving small sample size problems, they only obtain slight improvements in model's performance.

Due to the limitations of previous studies on processing inaccurately and uncertainly insufficient data, virtual sample generation (VSG) techniques, which enable to generate new samples arbitrarily, have been attracting lots of attention to solving the small sample size problems. In general, VSG techniques can be broken into three categories: (1) sampling-based VSG, typified by bootstrapping and synthetic minority over-sampling technique (SMOTE); (2) deep leaning-based VSG, represented by generative adversarial networks (GANs) and variational autoencoders (VAEs); (3) information diffusion-based VSG, including

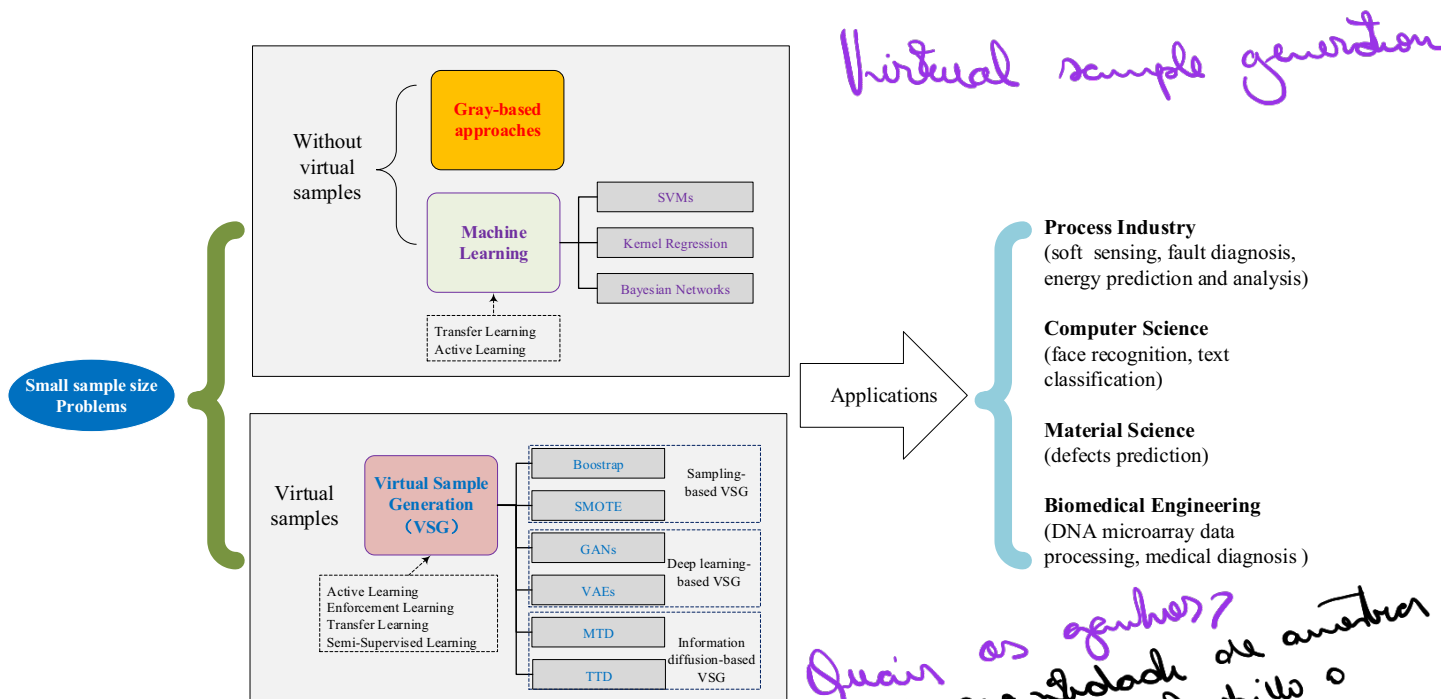


Fig. 1 The methods for solving small size problems

Virtual sample generation
 "Quais os ganhos?"
 "maior" quantidade de amostras
 mais representativas o
 conjunto de

Prever espaços esparsos
 → nem na informação não é adequada

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Présumpto?

mega trend diffusion (MTD) and tree-based trend diffusion (TTD). Sampling-based methods are much less expensive in computation than other methods. The performance of the methods depends on fitting a good distribution to match data through a parameterized probabilistic model. For high-dimensional data, formulating a joint distribution of these probabilistic models is difficult to fulfill. In order to reduce difficulties in parameter estimation for the probabilistic models, bootstrapping (Tsai and Li 2008) is used to generate copies of samples by over-sampling with replacement, but zero new information is carried with the generated samples. Instead of over-sampling with replacement, SMOTE generates synthetic example by randomly picking up data points along a line joining each minority class sample to any of the k -nearest neighbors (Chawla et al. 2002). Information carried with virtual samples by SMOTE is also very limited since any region beyond the line is not explored. Owing to the nonlinear multilayer structure, deep learning-based VSG methods like GANs (Gao et al. 2019; Tian et al. 2019) and VAEs (Huang et al. 2018) have more powerful in feature representation, which makes them particularly suitable for generating structural virtual samples (e.g., fake images) in the field of image recognition. Those deep generative models are usually difficult to train and prone to mode collapse, and their complex computations and expensive costs restrict their extensions to other fields. Considering adversarial and variational behavior, deep generative models become more complicated and time-consuming, which increases the difficulties in reliability analysis. Information diffusion-based VSG methods use a principle of information diffusion to derive a diffusion function and calculate virtual values to fill information gaps among points by discretizing the intervals of data. Compared with other two categories of VSG methods, they are simple and easy to understand. Most of them are combined with fuzzy theories, e.g., MTD (Li et al. 2007) and TTD (Li et al. 2012). One of the most important steps of information diffusion-based VSG methods is to determinate feature expansion range reasonably. In the work (Chen et al. 2017) and (Gong et al. 2017), the authors developed an information-expanded function based on triangular membership to determine feature expansion range and then optimized virtual samples by particle swarm optimization. For many other engineering applications, however, it is difficult to find an appropriate feature expansion range. The skewness of data has a great impact on the generation process of virtual sample with information diffusion-based VSG methods, which may violate the uniformity and consistency of data distribution.

The previous methods are either computationally expensive or pay more attention to the underlying distribution but ignore data feature space. In contrast to other methods, Kriging models provide a low computational cost

alternative to capture data feature (Bouhlef and Martins 2018). Meanwhile, we believe that a well-represented feature space of small datasets is supposed to cover data tendency. Based on these views, this study focuses on generating virtual samples in data sparse regions using a dimension-wise Kriging interpolation process. The main contribution of this study resides in that we generate virtual samples from feature space represented by a Kriging model instead of trying to fit its underlying distribution. In contrast to the previous methods, the obtained virtual samples by our Kriging-VSG method are thus more similar to the corresponding actual samples. The rest of this paper is organized as follows. Section 2 reviews the Kriging method. Section 3 describes the proposed method in a detailed manner. In Sect. 4, simulations and analysis on the benchmark functions and the HDPE application are conducted. The concluding remarks are summarized in Sect. 5.

2 Overview of Kriging

In geostatistics, Kriging is recognized as a common stochastically local spatial interpolation method. Variogram theory and structural analysis serve as a solid theoretical basis, which enables it to perform an unbiased estimation of the regionalized variables in the sense of minimum mean square error (Silva and Costa 2016). Kriging is also regarded as a semi-parametric interpolation method. Unlike traditional parametric interpolation methods (such as polynomial interpolation), Kriging does not need to select a fixed basis function, so it has more flexibilities. The research on Kriging has attracted a lot of attention. Several extensions of Kriging are designed based on gradient information and spatial cross-correction, e.g., gradient-enhanced Kriging (Bouhlef and Martins 2018). A Matlab toolbox for Kriging named ooDACE which provides some available implementations of Kriging and its extensions is introduced in the literature (Ulaganathan et al. 2015). Up to date, Kriging and its extensions have been widely used in the fields of engineering optimization, reliability analysis and computer simulation (Jamaly and Kleissl 2017; Silva and Costa 2016; Sun et al. 2017).

Considering a set of distinct points x_i of dimension d , $i = 1, 2, \dots, n$ and $x_i \in \mathbb{R}^d$, y_i denotes the observed values at known points x_i . In general, the Kriging method can be mathematically expressed as in the following formula:

$$f(x) = \mu(x) + z(x) \quad (1)$$

where $\mu(x)$ is a deterministic global approximation, and $z(x)$ is a stochastic Gaussian process with a mean of zero and a prior spatial covariance function. Suppose $z(x)$ is an intrinsically stationary process ($\mu(x) = \mu$), a covariance function between z_i and z_j is defined as

$$c(z_i, z_j) = \sigma^2 R(x_i, x_j) \quad (2)$$

where $R(x_i, x_j)$ is a well-defined correlation function. For many engineering problems, the Gaussian function with parameters is one of the most popular correlation functions since it is infinitely differentiable. It is related to the distance between two points, which can be written in the following form:

$$R(x_i, x_j; \theta) = \prod_{k=1}^d \exp(-\theta_k \|x_{ik} - x_{jk}\|^2) \quad (3)$$

To estimate a set of aforementioned unknown parameters, i.e., μ , σ^2 and θ , maximum likelihood estimation (MLE) is often used to maximize the log-likelihood function:

$$L(\mu, \sigma; \theta) = -\frac{1}{2} [n \ln(2\pi\sigma^2) + \ln(\det \mathbf{R}) + (\mathbf{Y} - \hat{\mathbf{Y}})^T \mathbf{R}^{-1} (\mathbf{Y} - \hat{\mathbf{Y}}) / \sigma^2] \quad (4)$$

So, we have

$$\hat{\mu} = (\mathbf{u}^T \mathbf{R}^{-1} \mathbf{u})^{-1} \mathbf{u}^T \mathbf{R}^{-1} \mathbf{Y} \quad (5)$$

$$\hat{\sigma}^2 = \frac{1}{n} (\mathbf{Y} - \hat{\mathbf{Y}})^T \mathbf{R}^{-1} (\mathbf{Y} - \hat{\mathbf{Y}}) \quad (6)$$

where $\mathbf{u}$ is an n -vector of ones, $\mathbf{R}$ is an $n \times n$ correlation matrix with $R_{ij} = R(x_i, x_j)$. Considering $\hat{\mathbf{Y}}$ is an unbiased estimation of $\mathbf{Y}$, $\hat{\mathbf{Y}} = \hat{\mu} \mathbf{u}$, the aforementioned formula (4) is reduced to a concentrated likelihood function that only depends on θ

$$L(\theta) = -\frac{1}{2} [n \ln(2\pi\sigma^2(\theta)) + \ln(\det \mathbf{R}(\theta))] \quad (7)$$

To the best of our knowledge, there is no analytical solution to estimate θ in the formula (7); consequently, it is necessary to use optimization algorithms to find θ so that maximizes the likelihood function (Bouhlef and Martins 2018). Finally, the best linear unbiased estimator of $f(x)$ at the unknown point x is given by

$$\hat{f}(x) = \hat{\mu}(x) + \mathbf{r}^T(x) \mathbf{R}^{-1} (\mathbf{Y} - \hat{\mathbf{Y}}) \quad (8)$$

where

$$\mathbf{r}^T(x) = [R(x, x_1), R(x, x_2), \dots, R(x, x_n)]$$

is a correlation vector between the unknown point x and the given points x_i .

3 The proposed Kriging-based VSG

Due to the insufficient number of representative samples in massive data, the distribution of such representative samples is very poor. In high-dimensional space, the sparse

region between representative samples forms “holes”, which makes it difficult to extract effective information and knowledge. A fisherman forms a grid-like structure by knotting fibers to repair the holes in fishing nets. Inspired by the fisherman's repairing process, we attempt to generate the most likely virtual samples through a similar patching process. During the fisherman's repairing process, the fisherman first fixes the other directions of the fishing nets with some predetermined fibers, and then draws the fibers from the remaining single direction, thereby interweaving the fibers in different directions into mesh structures. We regard the fishing nets as interpolation surfaces, and the nodes woven as virtual samples. The virtual sample generation process is essentially the fisherman's patching nets process. In the following subsection, we will describe the proposed virtual sample generation in a detailed manner.

Suppose that $\{s_1^m, s_2^m, \dots, s_n^m\}$ represents the projection points on the m -th dimension of the n points $x_i \in \mathbb{R}^d$ in the input space and is arranged in ascending order. The distance between the two consecutive projection points in the i -th interval can be expressed by the following equation:

$$\text{dist}(s_i^m, s_{i+1}^m) = \|s_{i+1}^m - s_i^m\|^2, \quad i = 1, 2, \dots, n-1; \quad m = 1, 2, \dots, d \quad (9)$$

Under the condition that without arousing any confusion, the following subsection notes $\text{dist}(s_i^m, s_{i+1}^m)$ as dist_i . Let avgDist denote the average distance between all projection points

$$\text{avgDist} = \frac{1}{n-1} \sum_{i=1}^{n-1} \text{dist}(s_i^m, s_{i+1}^m) \quad (10)$$

If the average distance between all projection points is not greater than the distance between two consecutive projection points, then the new data points $\tilde{x}_i^m$ are generated at the center of the two consecutive projection points on the m -th dimension, i.e.,

$$\tilde{x}_i^m = \frac{1}{2} (s_i^m, s_{i+1}^m), \quad \text{if } \text{avgDist} \leq \text{dist}_i \quad (11)$$

Let Q_α^m denote α -quantile of the projection points, satisfying that $P(s^m \leq Q_\alpha^m) = \alpha$. For all dimensions excluding the m -th dimension, $q = \{1, 2, \dots, d\} \cap q \neq m$, each dimension is fixed to a specific quantile

$$\tilde{x}_i^q = Q_\alpha^q \quad (12)$$

Having combined the new data points across all dimensions, virtual samples can be generated in the input space:

$$\tilde{x}_i^k = \begin{cases} \frac{1}{2} (s_i^m, s_{i+1}^m), & \text{if } k = m \text{ and } \text{avgDist} \leq \text{dist}_i \\ Q_\alpha^q, & \text{if } k = q \text{ and } q = \{1, 2, \dots, d\} \cap q \neq m \end{cases} \quad (13)$$

vetor de correlação entre nova amostra e cada amostra existente

The output of virtual samples $\tilde{x}_i$ generated in the input space can be determined in the output space using the pre-established Kriging approximation model based on the original dataset $\{(x_i, y_i) | x_i \in \mathbf{X}, y_i \in \mathbf{Y}\}$. The pseudocode of the proposed method is described in Table 1.

Let $|Q_\alpha|$ denote the number of quantiles Q_α at different cumulative probabilities α . Since the new data point on the other dimensions instead of the m -th dimension is fixed to various quantiles, the appearance of these new data points

leads to a permutation and combination problem. Regardless of the m -th dimension, there are $|Q_\alpha|^{d-1}$ potential combinations of these new data points. On the m -th dimension, at most $(n-1)$ sparse intervals occur. Therefore, the proposed approach can generate not more than $(n-1)d|Q_\alpha|^{d-1}$ virtual samples. In order to make the generated virtual samples cover the entire input space to the maximum extent, in general, one can pick up α from $\{0.025, 0.25, 0.5, 0.75, 0.975\}$ so that Q_α can completely

Table 1 The pseudocode of the proposed Kriging-based VSG

Input: actual samples $\{(x, y) x \in \mathbb{R}^d, y \in \mathbb{R}\}$
Output: virtual samples $\{(\tilde{x}, \tilde{y}) x \in \mathbb{R}^d, y \in \mathbb{R}\}$
compute a projection s of each actual sample along each dimension and arrange them in ascending order.
construct a Kriging model with actual samples.
for each dimension $m = 1, 2, \dots, d$ do // compute the input part of virtual samples $\tilde{x}$.
compute $avgDist$ // the average distance between all projection points on dimension m .
compute Q_α^m // the α -quantile of the projection points, $\alpha \in \{0.025, 0.25, 0.5, 0.75, 0.975\}$.
for each interval between two consecutive projection points $i = 1, 2, \dots, n-1$ do
compute $dist_i$ // the distance between two consecutive projection points
if $avgDist \leq dist_i$ // the interval i is sparse region
$\tilde{x}_i^m \leftarrow (s_i^m, s_{i+1}^m) / 2$
end if
for each dimension except the m -th dimension, $q = \{1, 2, \dots, d\} \cap q \neq m$ do
$\tilde{x}_i^q \leftarrow Q_\alpha^q$ // α -quantile of the projection points for dimension q
end for
end for
compute the output part of virtual samples $\tilde{y}$ using the Kriging model.

capture the underlying distribution from the original data points.

4 Simulation results and analysis

In this section, a series of validation simulations are carried out to evaluate the effectiveness of the proposed method. Without loss of generality, validation simulations start by applying to two numerical simulations, and then, the proposed approach is extended to a real-world application from a cascade reaction process for high-density polyethylene. The computing environment is Intel(R) Core (TM) i7-6700 3.4 GHz 16 GB RAM with MATLAB R2018b installed. The MATLAB version of Kriging toolbox is required, which can be download from Design and Analysis of Computer Experiments (DACE) website at free.¹

4.1 Numerical simulations

Considering the following benchmark function with two input variables

$$f(x_1, x_2) = 1.3356(1.5(1 - x_1)) + e^{2x_1-1} \sin(3\pi(x_1 - 0.6)^2) + e^{3(x_2-0.5)} \sin(4\pi(x_2 - 0.9)^2) \quad (14)$$

where x_1 and x_2 are mutually independent; $\mathbf{x} = [x_1, x_2]^T$ is subject to stand uniform distribution with $\mathbf{x} \sim U(0, 1)$.

An original dataset with 30 data points was selected to establish an approximation model for fitting the benchmark function (see Table 2). The proposed method that described in Sect. 3 is applied to this two-dimensional numerical example. For the purpose of explanation clarity, we describe the generation process of virtual samples step by step. To begin with, the average distance for all sorted projection points on the x_1 -dimension is calculated by Eq. (10), and we obtain $avgDist = 0.0547$. Meanwhile, the distance of the two consecutive projection points is calculated by Eq. (9). For instance, taking the interval between the 4-th projection point and the 5-th one into consideration, we find $dist_4 = 0.0769$. Obviously, the 4-th interval on the x_1 -dimension is recognized as a sparse interval because $dist_4 > avgDist$. Therefore, a new data point is generated at the center of the sparse interval. The projection of the new data point on x_1 -dimension is $G = (s_4 + s_5)/2 = 0.1156$. Fixing x_1 -dimension to G , the surface of $z = f(x_1, x_2)$ reduces to the curve of $y = f(x_2)|_{x_1=G}$. We specify x_2 -dimension as different

quantiles varying from $Q_{0.025}$, $Q_{0.25}$, $Q_{0.5}$ and $Q_{0.75}$ to $Q_{0.975}$. Combining G with this quantiles, we can obtain five combinations: $(G, Q_{0.025})$, $(G, Q_{0.25})$, $(G, Q_{0.5})$, $(G, Q_{0.925})$ and $(G, Q_{0.95})$, illustrated in Fig. 2. Repeating the aforementioned steps for all of the sparse intervals on x_2 -dimension, we can obtain 135 virtual samples from the surface $z = f(x_1, x_2)$. Finally, the corresponding output values of the virtual samples are computed by the Kriging approximation model. The surface of the benchmark function is shown in Fig. 3.

Since the approximation model constructed on the original dataset is critical to the goodness of the virtual samples generated by the proposed method, it is necessary to investigate the performance of the approximation model. To construct a reasonable approximation model, three popular interpolation methods: Kriging, radial basis function (RBF) interpolation, k -nearest neighbors algorithm (k -NN) and two common regression methods (i.e., ELM and Bootstrap-ELM) are compared. The average values of root-mean-square error (RMSE) and mean absolute error (MAE) are used to evaluate the accuracy of different approximation models in leave-one-out cross-validation (LOOCV) tests. The definitions of these measures are given as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (15)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (16)$$

where $\hat{y}_i$ denotes the estimated values; y_i denotes the observed values. For the purpose of fair comparisons, proper parameters for different approximation models should be carefully configured. In this simulation, genetic algorithm (GA) optimization is used to find the parameters θ of the Kriging model. The concentrated likelihood function expressed in Eq. (7) achieved maximum as the solution for the parameters θ of the Kriging model equals to $\theta = [0.1790, 0.1748]^T$. For RBF interpolation, we use the Gaussian radial basis function. The selection of the shape parameter σ in the Gaussian radial basis function is based on the rule of thumb, i.e., $\sigma = d_{\max}/\sqrt{2n} = 0.1714$, where $d_{\max}$ is the maximum distance between the given data points of $n = 30$. On the basis of numerous LOOCV tests, the size of the neighbors in k -NN and the number of the neurons in ELM are assigned to 2 and 12, respectively. In Bootstrap-ELM, 50 bootstrap samples with the size of $n = 30$ are obtained by resampling with replacement from the original samples to train individual ELMs. An aggregated model is consequently constructed by assembling the output of individual ELMs. Table 3 shows the testing errors on the obtained 135 virtual samples for different

¹ http://www2.imm.dtu.dk/pubdb/views/publication_details.php?id=1460.

Table 2 The selected data points for the benchmark function $f(x_1, x_2)$

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	x_1	x_2	$f(x_1, x_2)$		x_1	x_2	$f(x_1, x_2)$		
Training data points	1	0.4522	0.3192	0.7824	Testing data points	1	0.8492	0.3298	0.9312
	2	0.3904	0.2042	1.4808		2	0.7384	0.7672	1.3110
	3	0.9764	0.0700	2.7607		3	0.4299	0.1582	1.6050
	4	0.5233	0.9500	1.1487		4	0.2072	0.2864	1.6383
	5	0.1109	0.1412	2.4417		5	0.3234	0.6871	2.7847
	6	0.3299	0.7213	2.5728		6	0.3752	0.5121	2.6119
	7	0.3421	0.9288	1.8034		7	0.8171	0.7321	1.8783
	8	0.5211	0.4073	1.1039		8	0.5317	0.7498	1.5906
	9	0.1203	0.5209	3.2106		9	0.7743	0.2395	0.6192
	10	0.6255	0.2191	0.5796		10	0.3346	0.6629	2.8536
	11	0.3466	0.8424	1.8635		11	0.5746	0.8162	1.0994
	12	0.1992	0.9869	2.5832		12	0.8639	0.7939	1.8812
	13	0.2983	0.7699	2.4064		13	0.1986	0.4691	2.8353
	14	0.8899	0.7061	2.6210		14	0.6725	0.3095	0.2018
	15	0.5014	0.5953	2.3293		15	0.9018	0.6876	2.8333
	16	0.2770	0.7529	2.5767		16	0.4965	0.8296	1.2912
	17	0.5340	0.4967	1.8732		17	0.4128	0.0680	1.6482
	18	0.5743	0.8651	0.9187		18	0.2343	0.6137	3.3226
	19	0.0148	0.9685	2.2113		19	0.7750	0.8715	0.9821
	20	0.7028	0.0988	1.0467		20	0.9122	0.7888	2.3591
	21	0.5067	0.5470	2.2373					
	22	0.3813	0.4030	1.6296					
	23	0.0649	0.1070	2.3883					
	24	0.3586	0.7242	2.4399					
	25	0.2035	0.7830	2.5703					
	26	0.8138	0.5666	2.3635					
	27	0.3934	0.8113	1.8007					
	28	0.0536	0.5768	3.2736					
	29	0.3751	0.9440	1.7206					
	30	0.1653	0.5076	3.1540					

approximation models. From Table 3, it can be found that the Kriging model has a best performance (are highlighted in bold), followed by RBF interpolation. k -NN is moderate, whereas both ELM and Bootstrap-ELM have much larger testing errors. On average, Bootstrap-ELM is slightly better than ELM.

As exhibited in Fig. 2, the projection points of the virtual samples generated by the proposed method are fully covered in the entire input space. One can easily find that the information gaps between the original samples are dramatically reduced. Due to the good approximation ability, we use the Kriging model to estimate the output of the benchmark function at the generated data points, shown in Fig. 4. The 135 virtual samples whose output values are estimated by the Kriging model are drawn in Fig. 3. From those simulation results demonstrated by Fig. 2 and Fig. 3,

it can be seen that all of the newly generated virtual samples fall down into the scope of the expected sparse regions so that make the distribution of samples more uniform. It means that the Kriging model has an ability to reflect the behavior of inherent mechanism of an unknown system.

To further demonstrate the effectiveness of the proposed Kriging-VSG method, we tested it on a benchmark function with three input variables, which can be written as

$$h(x_1, x_2, x_3) = 4(x_1 - 0.5)(x_2 - 0.5) \sin\left(2\pi\sqrt{x_2^2 + x_3^2}\right) \quad (17)$$

where x_i 's are mutually independent, and $\mathbf{x} = [x_1, x_2, x_3]^T \sim U(-1, 1)$. We picked up 50 data points from the benchmark function, 30 out of which were used as

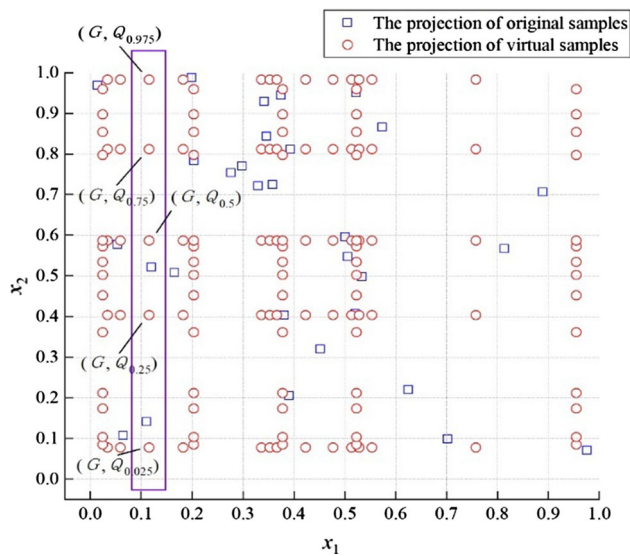


Fig. 2 The two-dimensional projections of the original and virtual samples

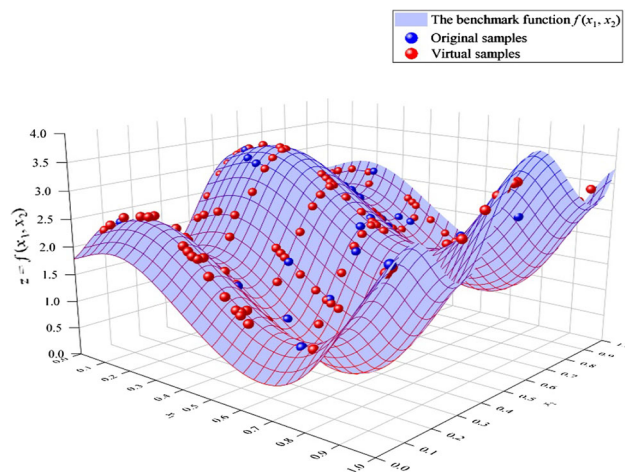


Fig. 3 The surface of the benchmark function $z = f(x_1, x_2)$

the training dataset and the remaining as the testing dataset, shown in Table 4. The generation procedure of virtual samples was conducted, and 268 virtual samples were obtained. For our methods, the parameters for gaussian function $\theta = [0.3961, 0.3939, 0.4006]^T$. The parameters of other methods are configured below: the shape parameter for RBF interpolation $\sigma = 0.3746$; the size of neighbors for k -NN is 8; the number of the neurons in ELM or Bootstrap-ELM is 20, and the remaining settings in Bootstrap-ELM is the same as the first simulation. The testing errors of

different approximation models are shown in Table 5. From Table 5, it is observed that ELM has the largest testing errors while both the values of average RMSE and average MAE of our method are much smaller than those of other methods. Bootstrap-ELM is slightly better than ELM. This result is consistent with that obtained in the first simulation. The estimated output values of virtual samples by our method (the best model) are exhibited in Fig. 5.

4.2 Real-world application: high-density polyethylene process

High-density polyethylene (HDPE) is a kind of thermoplastic resin with high crystallinity and non-polarity formed by ethylene copolymerization. It is widely used in film production, blow molding, pipe and other applications. The HDPE process is mainly composed of four parts, namely raw materials and catalyst pretreatment section, polymerization reaction section, subsequent treatment section and hexane solvent recycling section, in which the polymerization reaction is a quite important step. The polymerization reaction section is essentially concerned with the polymerization of high-purity ethylene monomer dissolved in hexane slurry to form high-density polyethylene under certain temperature and pressure conditions. This section comprises two polymerization reactors connected with flash tanks in cascade, a post-reactor and a recycle unit. Figure 6 shows the flowchart of HDPE cascade reaction.

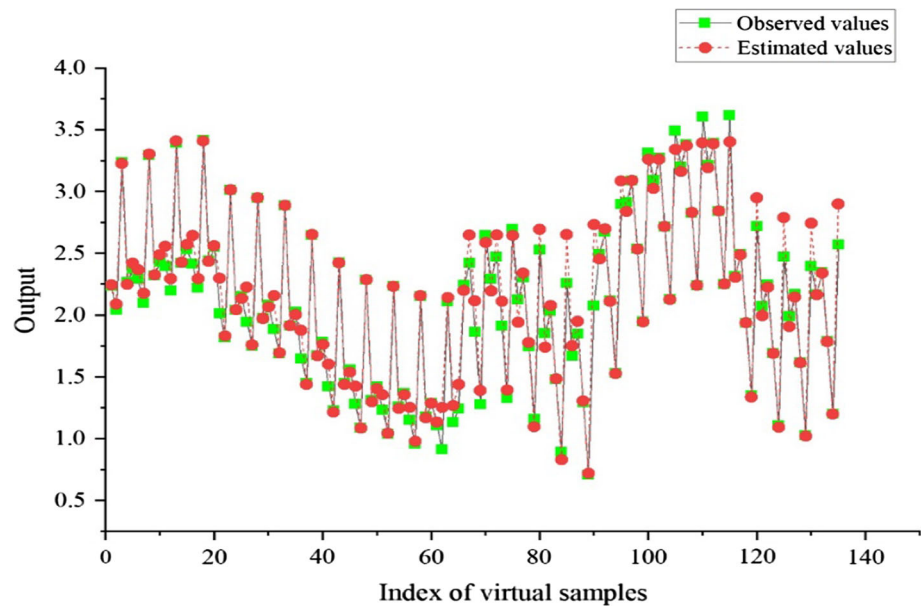
The melting index MI is one of the most important quality indexes in the production of HDPE. Because on-line measurement methods of MI are not available, off-line laboratory analysis is carried out to estimate the quality indicator. Therefore, time delay during the off-line phase directly affects product quality examination as well as operation on-line optimization. On the basis of process mechanism analysis and accumulated practical experience, MI in reactor A depends mainly on the seven process variables listed in Table 6.

To develop a reasonable model of soft sensing of MI in production quality examination in HDPE cascade reaction, the seven process variables summarized in Table 6 are selected as the input variables, and the melting index in reactor A is taken as the output variable. A set of instances with seven inputs and one output are collected from the historical database stored in the distributed control system for HDPE. The sampling interval between instances is

Table 3 Testing errors on virtual samples for different approximation models

	Kriging	RBF interpolation	k -NN	ELM	Bootstrap-ELM
Average RMSE	0.1262	0.2697	0.4489	0.6099	0.5155
Average MAE	0.0703	0.1536	0.3259	0.4735	0.4021

Fig. 4 The output of the benchmark function $f(x_1, x_2)$ at the virtual data points



10 min. On the collected dataset, the data-preprocessing technology (e.g., data fusion, data filtering, outlier detection and removal, steady-state identification) is applied to eliminate noise, inconsistent and missing data. Although the above data-preprocessing is performed, the dataset also contains a lot of redundant data. To reduce redundant data, we use random selection with selection ratio 1/6 to remove redundant instances. We obtain a small dataset with only 104 representative samples, in which two-thirds of the samples are used as the training dataset, and one-third of the samples are used as the testing dataset.

We use ELM to learn the relationship between MI and the seven selected variables based on the training dataset. By using the trial-and-error method, the optimal topology of ELM is determined based on LOOCV. The testing errors of ELMs with different hidden neurons are depicted in Fig. 7. We can find the optimal number of hidden neurons of ELM is 14 with respect to average values of $RMSE$ and MAE . As shown in the second column Table 7, without any virtual samples, the obtained ELM has a relatively large testing error, i.e., average $RMSE = 18.1813$, average $MAE = 15.2059$. In order to further improve the accuracy of the obtained ELM models, 184 virtual samples are generated by the proposed method to enlarge the original training dataset. Due to the slight fluctuation of process variables under normal operating conditions of HDPE, in the proposed method, we set α as 0.5 instead of $\{0.025, 0.25, 0.50, 0.75, 0.975\}$ to reduce computational cost caused by the permutation and combination problem described in Sect. 3. The parameters $\theta = [1.66, 0.78, 0.92, 0.53, 0.39, 0.45, 0.53]^T$ of the Kriging model is determined by GA optimization to

maximize the concentrated likelihood function expressed in Eq. (7). Then, a new training dataset consisting of 253 samples is obtained by adding the generated 184 virtual samples to the original 69 training samples. Another ELM with hidden neurons of 14 is used to learn on the new training dataset and to be evaluated on the same testing dataset. The solid red line and dash green line in Fig. 8 show the predicted output values of ELMs on testing samples with and without adding virtual samples by the proposed method, respectively.

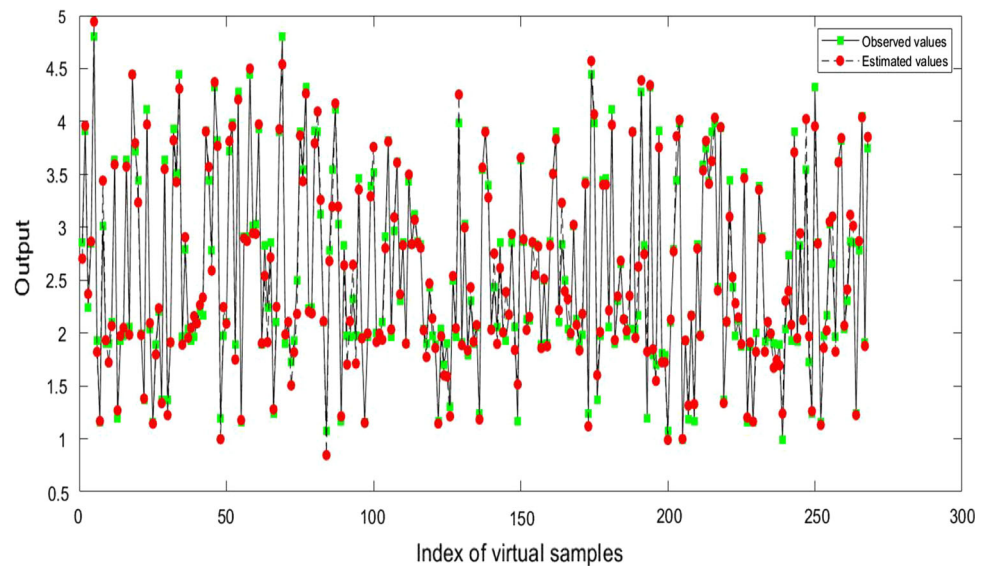
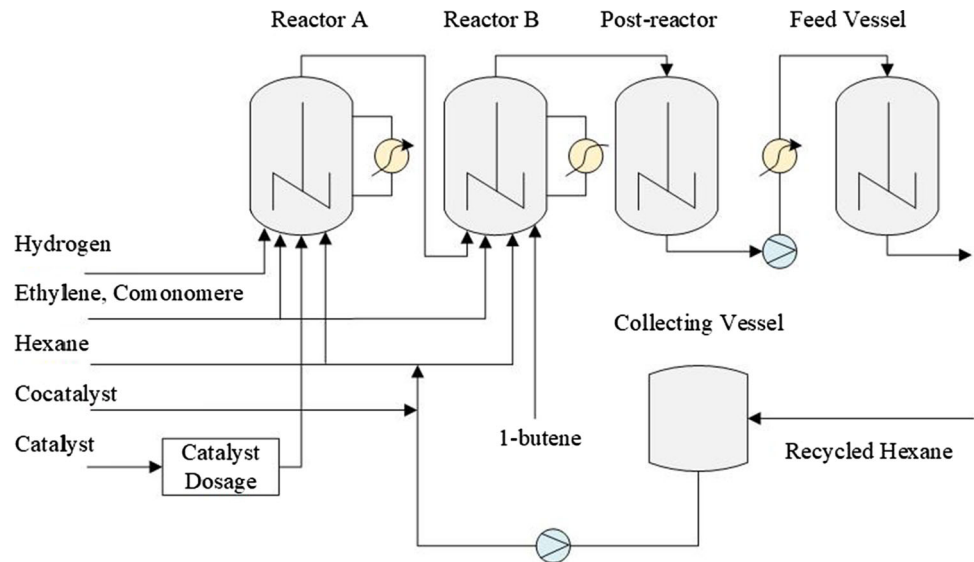
To demonstrate the effectiveness and superiority of the proposed method, bootstrap and other VSG methods including MTD and TTD are compared with the proposed method. It is worth mentioning that bootstrap takes repetition of actual samples from the original training dataset with replacement instead of generating new samples. Table 7 shows the comparison results of testing errors of ELMs for different methods. Table 8 show the stability of different models under 50 rounds of LOOCV. In Table 7, regardless of MTD and TTD, or Bootstrap, their performance is greater than that of the proposed method in terms of average $RMSE$ and average MAE . All of them are obviously better than that without any virtual sample. It can be concluded that virtual samples contribute to improvement on the prediction ability of forecasting models. The best performance we obtained is the average $RMSE$ of 13.2011 with the average MAE of 10.6914. From Table 8, although MTD and Bootstrap have large testing errors, they show good stability. Still, their stability is far less stable than our method.

Table 4 The selected data points for the benchmark function $h(x_1, x_2, x_3)$

Training data points			Testing data points		
1	- 0.659136	0.331974	- 0.289853	0.283575	0.975869
2	- 0.206401	0.033116	- 0.551657	- 0.428498	- 0.484415
3	- 0.195223	0.906914	- 0.225509	0.452467	- 0.85201
4	0.241344	- 0.926874	- 0.157775	- 0.541672	0.368192
5	- 0.69126	0.618408	- 0.631799	0.375562	0.96567
6	- 0.23731	0.497238	0.451551	- 0.00718	- 0.195632
7	- 0.677732	- 0.759626	- 0.259275	- 5.612204	0.371071
8	0.516225	0.05009	0.68312	0.026794	0.664847
9	0.742222	- 0.348333	0.468459	0.412931	0.19498
10	- 0.298447	0.092899	0.142052	1.138312	- 0.329377
11	- 0.411703	- 0.169813	0.914768	- 1.034538	- 0.40155
12	0.061259	- 0.638524	- 0.469356	- 1.927354	- 0.094815
13	- 0.154709	- 0.672975	- 0.638766	- 1.345232	- 0.280787
14	0.116638	0.589316	0.446347	0.13665	- 0.15133
15	0.485091	0.154788	- 0.305125	0.017233	- 0.419629
16	- 0.141288	- 0.484773	- 0.232263	- 0.590339	- 0.519043
17	- 0.750254	0.503893	0.254693	0.007687	0.527796
18	- 0.951132	- 0.542661	- 0.9567	3.553963	0.481296
19	- 0.364959	0.534659	0.601117	0.112955	- 0.788159
20	0.30738	0.342404	0.491695	- 0.070858	- 0.649981
21	0.913872	0.430425	0.626226	0.114958	
22	0.871462	0.284122	- 0.233387	- 0.23699	
23	- 0.084227	- 0.161903	0.234558	1.509536	
24	0.518655	- 0.365144	- 0.44986	0.030887	
25	0.487377	0.578147	- 0.096722	0.002034	
26	0.363121	0.011273	0.608899	- 0.169271	
27	- 0.073479	0.271323	0.972208	0.030829	
28	- 0.575674	0.901789	- 0.940016	- 1.635099	
29	- 0.802963	- 0.112072	0.071328	2.364091	
30	0.647149	- 0.879962	- 0.825846	- 0.782495	
					0.262377
					0.994007
					0.304902
					0.209981
					- 0.715626
					- 0.94973
					- 0.64629
					0.849162
					- 0.552459
					- 0.252872
					- 0.824999
					0.280233
					- 0.909898
					0.321234
					0.82114
					0.15099
					0.060103
					- 0.502742
					- 0.544574
					0.604183
					0.330021
					2.198559
					- 1.129067
					- 0.023345
					- 0.620481
					0.765491
					0.037979
					- 0.324917
					0.081715
					1.105821
					0.765967
					- 0.474156
					0.468026
					- 1.066548
					1.347744
					4.773505
					2.914621
					- 0.01104
					0.00908
					0.6694

Table 5 Testing errors on virtual samples for different approximation models

	Kriging	RBF interpolation	k -NN	ELM	Bootstrap-ELM
Average <i>RMSE</i>	3.2699	5.491	5.5377	6.1736	6.0858
Average <i>MAE</i>	2.7887	4.3033	4.9239	4.5323	4.1364

Fig. 5 The output of the benchmark function $h(x_1, x_2, x_3)$ at the virtual data points**Fig. 6** The process flow diagram of HDPE cascade reaction**Table 6** The process variables which have an influence on *MI*

Variable no.	Tag name	Description
1	TRC10403	The temperature of reactor A
2	PR10403	The pressure of reactor A
3	FRC10301/2	The feed flow of catalyst in reactor A
4	FRC10405/6	The feed flow of hydrogen in reactor A
5	FRC10409	The feed flow of high-purity ethylene in reactor A
6	PC2A	The partial pressure of the ethylene in reactor A
7	H2C2A	The partial pressure ratio of ethylene to hydrogen in reactor A

Fig. 7 The testing errors of ELMs against the number of hidden neurons

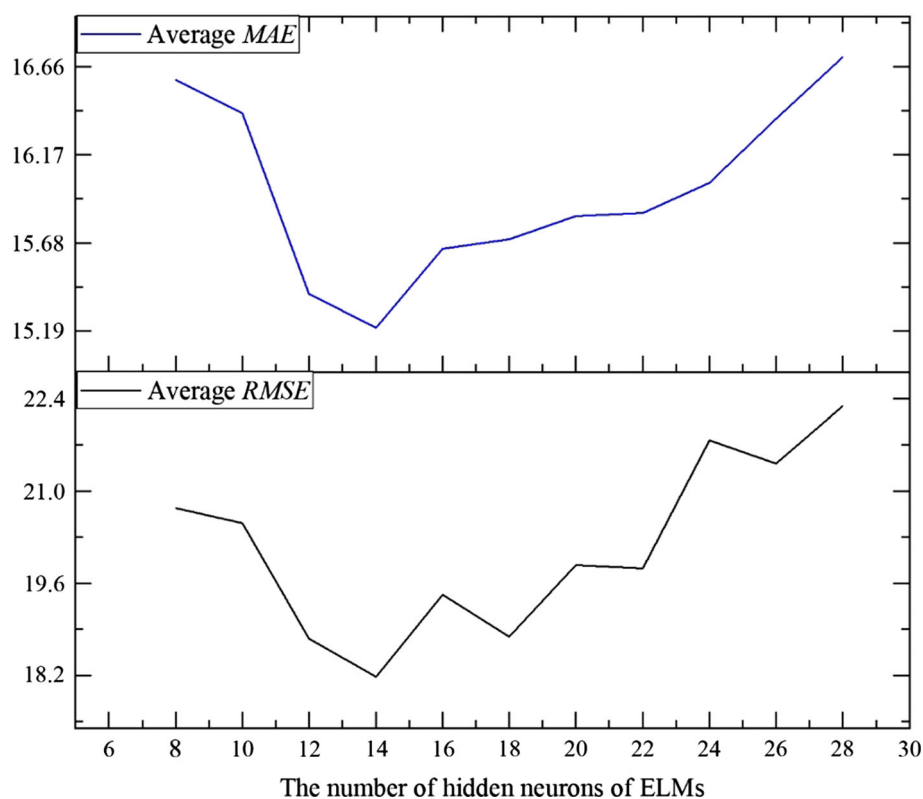


Table 7 The comparisons of testing errors for different methods

	The proposed method	Without VS	MTD	TTD	Bootstrap
Average <i>RMSE</i>	13.2011	18.1831	15.9662	16.2193	16.9145
Average <i>MAE</i>	10.6914	15.2059	12.1285	13.5196	14.4873

Fig. 8 The predicted output of testing samples

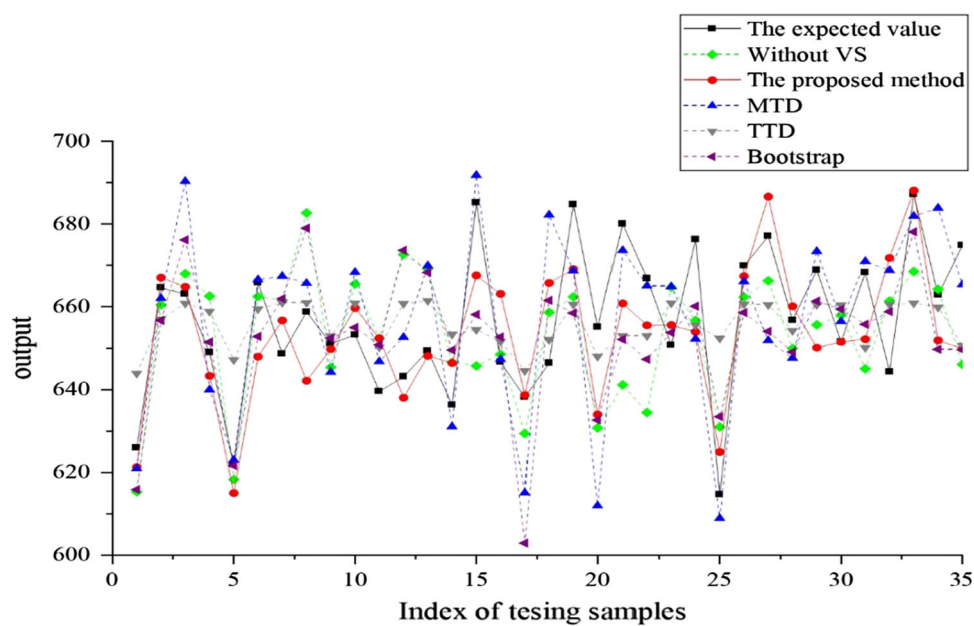


Table 8 The statistical description for 50 rounds of LOOCV

	The proposed method	Without VS	MTD	TTD	Bootstrap
Std. of average <i>RMSE</i>	2.2562	8.2123	4.8545	6.4548	4.6631
Std. of average <i>MAE</i>	2.1311	8.0041	4.6326	6.1029	4.2445

5 Conclusion and future direction

In the modern process industries, data-driven methods are used to construct prediction models. The prediction accuracy of constructed models depends on the number of effective samples; therefore, data-driven methods will not work well in uncertain small sample size environments. In order to obtain enough representative samples for enhancing the generalization ability of the existing data-driven methods, a novel approach named Kriging-VSG is proposed to generate feasible virtual samples. Taking the reduction of information gaps into consideration, the proposed Kriging-VSG is used to generate virtual samples in data sparse regions through a dimension-wise interpolation process. The data sparse regions are founded according to a well-defined distance-based criterion. The effectiveness of the proposed Kriging-VSG is proven by two numerical simulations and a real-world application (HDPE). The results indicate that the proposed method outperforms other existing approaches such as MTD, TTD and Bootstrap.

For the future research, the moving window on nearest neighbors of original samples can be considered for obtaining local tendency. As in our experience, such local information is critical for capturing the feature space of small sample size problems. Therefore, the more qualified virtual samples can be derived from the feature space. In addition, sample entropy-based method can also be applied to find the optimal number of the virtual samples on the moving window.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants performed by any of the authors.

Informed consent No individual participants are included in the study.

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